Class 12 - Important Formulas

Chapter 13 - Nuclei

S.No.	Term	Description
1	**Isotopes**	Nuclei having same atomic number but different mass number.
2	**Isobars**	Nuclei having different atomic number but same mass number.
3	**Atomic mass unit**	One atomic mass unit is defined as one twelfth part of the mass of $^{12}C_6$ atom. Value of a mass unit is $1u=1.66054\times10^{-27}Kg=931$ MeV
4	**Nuclear radius**	$R=R_0A^{1/3}$ Where value of $R_0 \cong 1.2 \times 10^{-15} \cong 1.2$ fm and is known as nuclear radius parameter
5	**Nuclear density**	The density of nuclear matter is approximately of the order of 10^{17} Kg/m^3 and is very large compared to the density of ordinary matter.
6	**Nuclear forces**	It is the force which holds the nucleons together inside the nucleus.
7	**Mass defect**	$\Delta m = [Zm_p + (A-Z)m_n] - m$
8	**Binding energy**	$BE = \Delta mc^2 = [[Zm_p + (A-Z)m_n] - m]c^2$
9	**Binding energy per nucleon**	=BE/A
10	**Radioactive decay law**	$N = N_0 e^{-\lambda t}$
11	**Half life**	$T=\ln 2/\lambda$
12	**Average life**	$T_{av}=1/\lambda$